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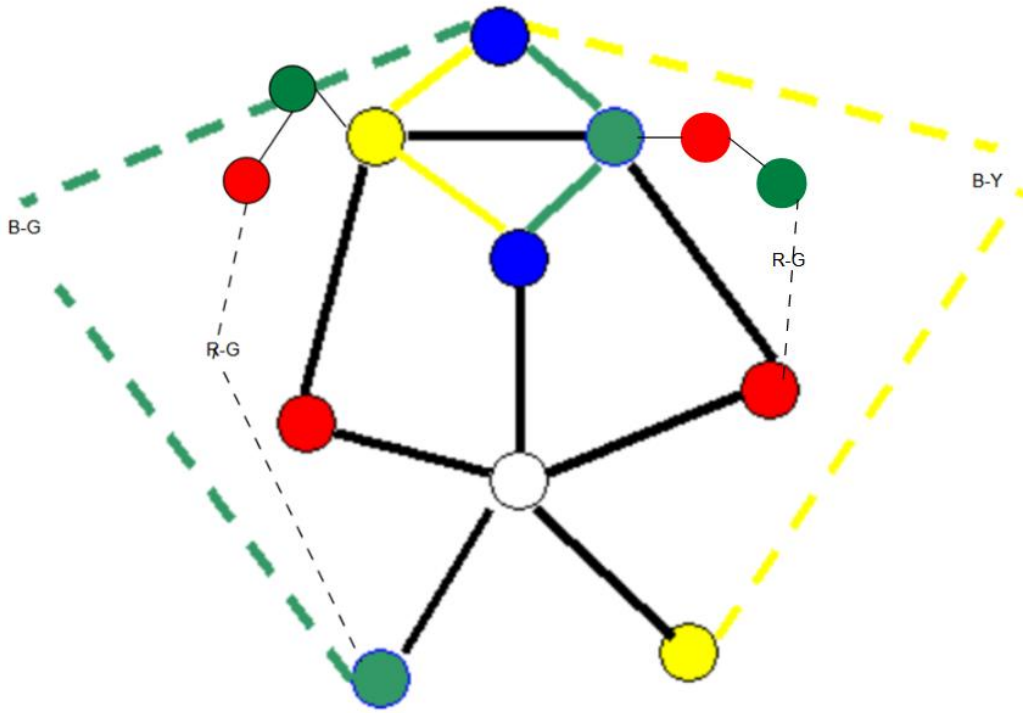
This document shows that the following graph generates a counterexample to Kempe's 4 colour theorem:



- The R-G Kempe chain on the left does not need to intersect the B-G kempe chain.

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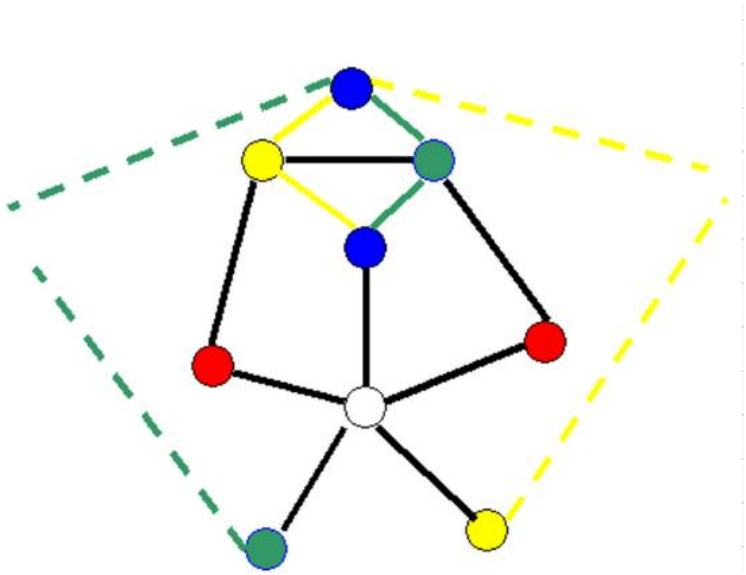
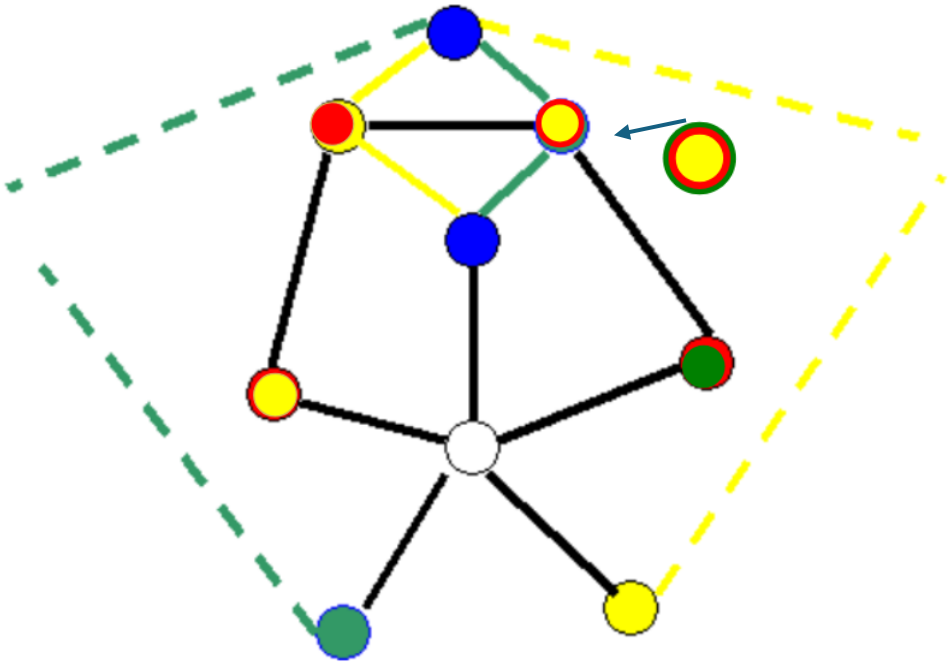
<u>Heawood's Counterexample from Shai Simonson:</u>



In our seminar, Heawood's counterexample could be made 4-colourable with a coloring like the one below. The node on the right has a Green-Red-Yellow coloring in that chronological order.



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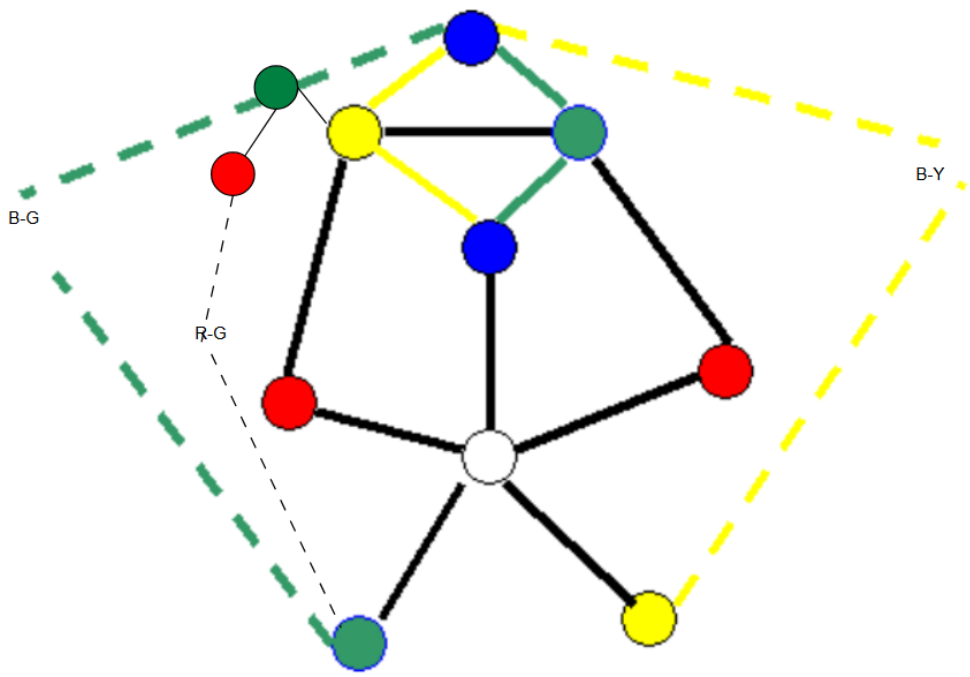
Forming a counterexample:

The process starts off from the beginning with the R-G Kempe chain on the left added. A counterexample will be eventually generated.

→Assume an R-G Kempe chain connected from Y in the original graph. The R-G Kempe chain can intersect the B-G Kempe chain 0,1, or 2 times.



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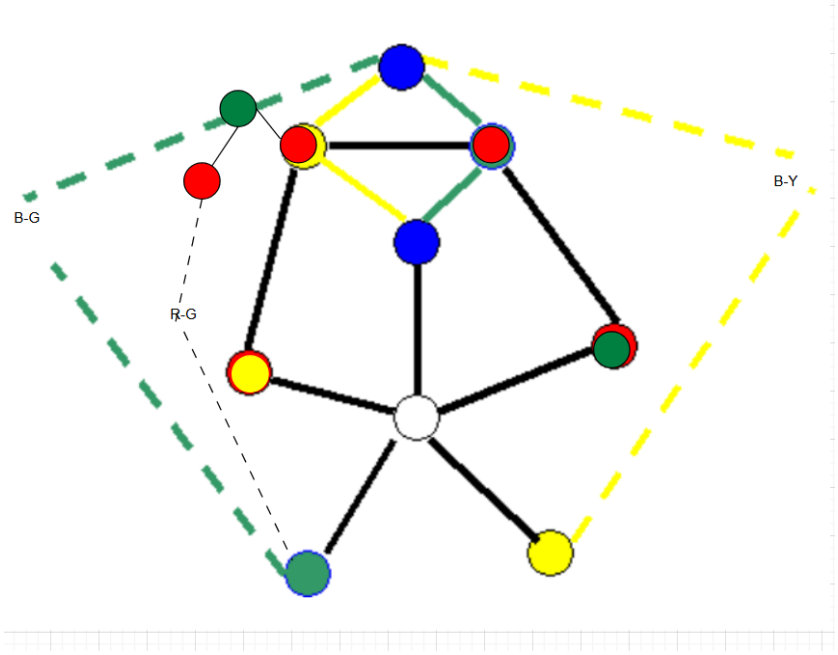
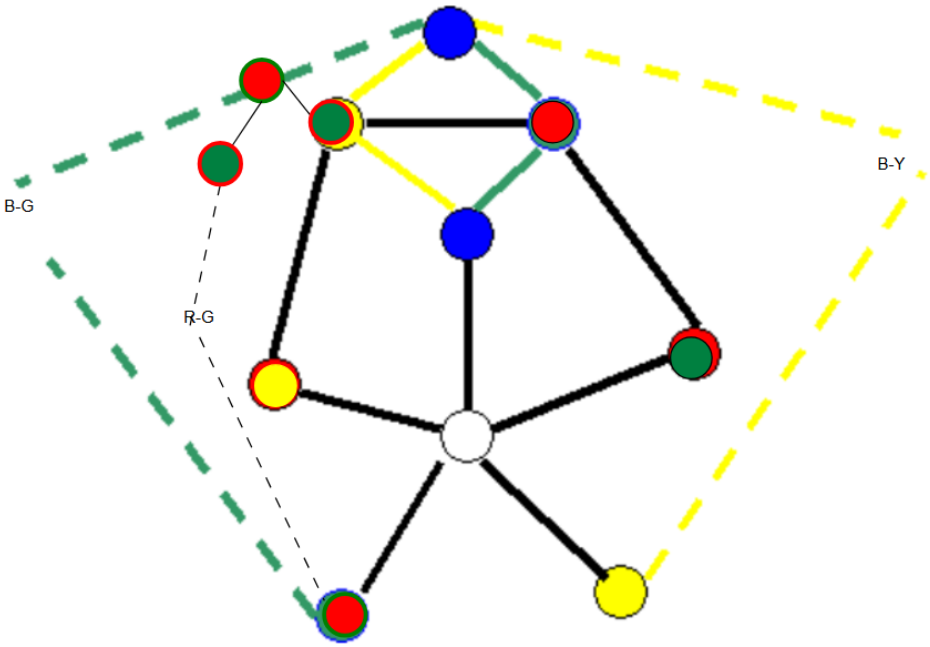
→Begin from the original Kempe Method by toggling the two R-Y and R-G pairs.



→Toggle the R-G Kempe chain to remove the adjacent red coloured nodes.



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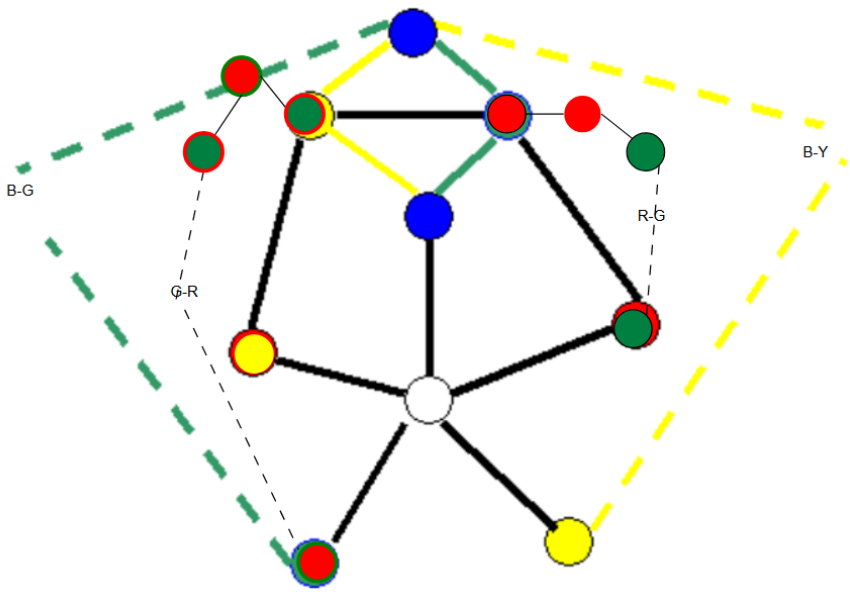
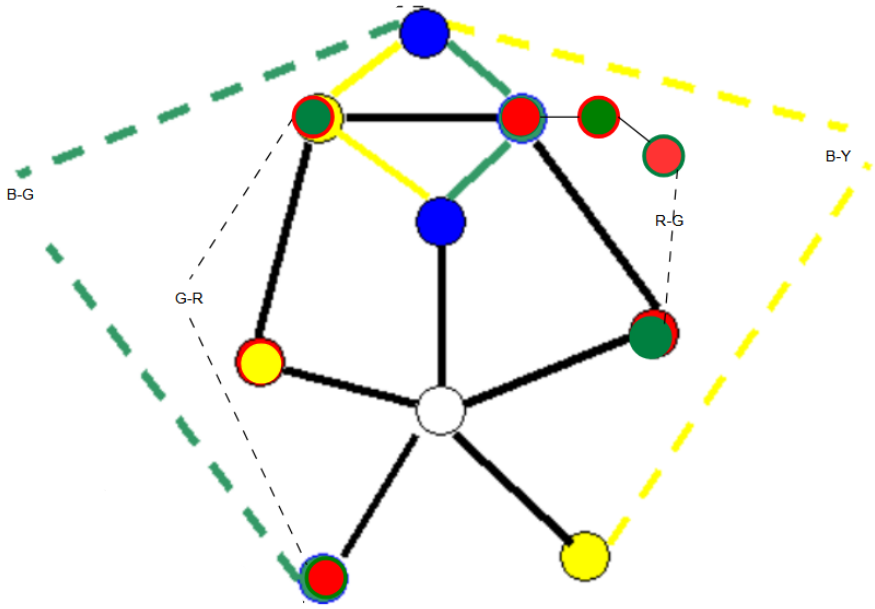
→Assume an R-G Kempe chain was on the right side of the original graph as well.



→Toggle the added R-G network / Kempe chain, excluding the existing connected R-G nodes to make the adjacent nodes different coloured. The R-G Kempe chain on the is now made simplified.



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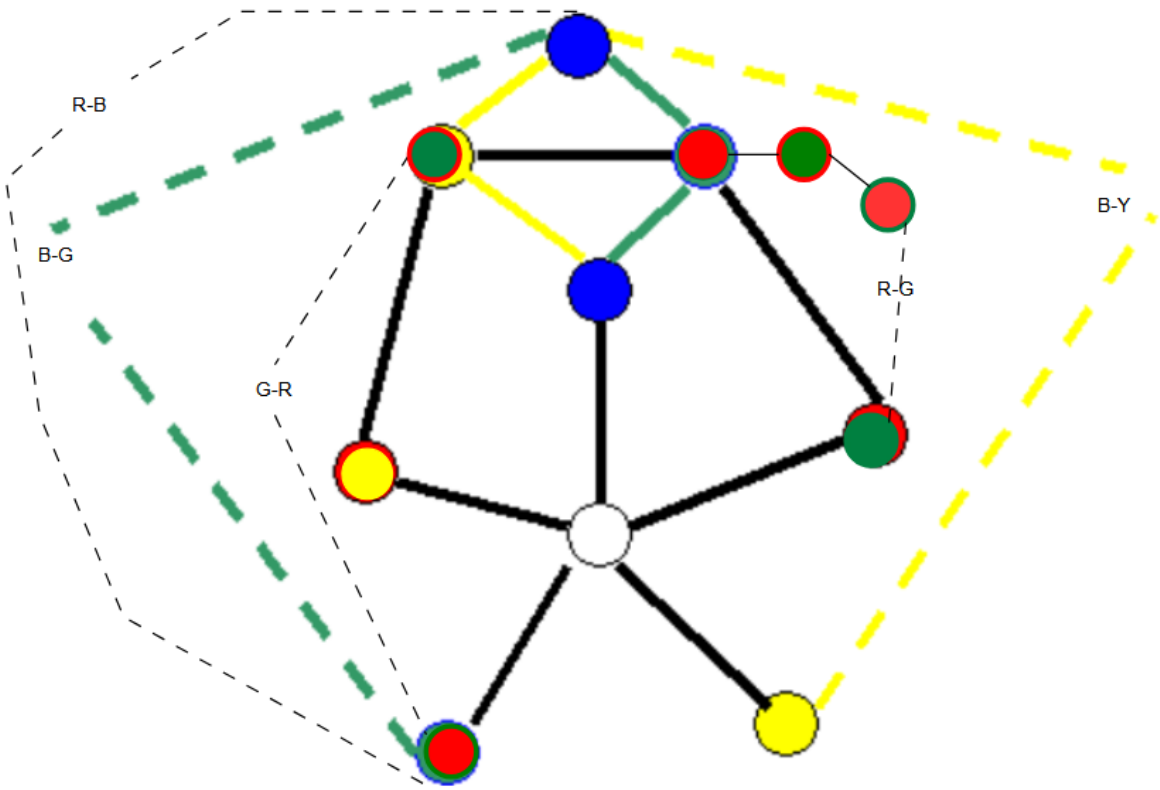
This results in all the neighbours of the uncoloured node still being all 4 colours. Where we left off in the seminar.

Following Kempe's method again:

1)

A. Attempt to make the bottom left red node Blue. This fails due to a potential R-B Kempe chain (newly added).

B. Attempt to make the bottom left red node Green. This fails due to R-G Kempe chains.



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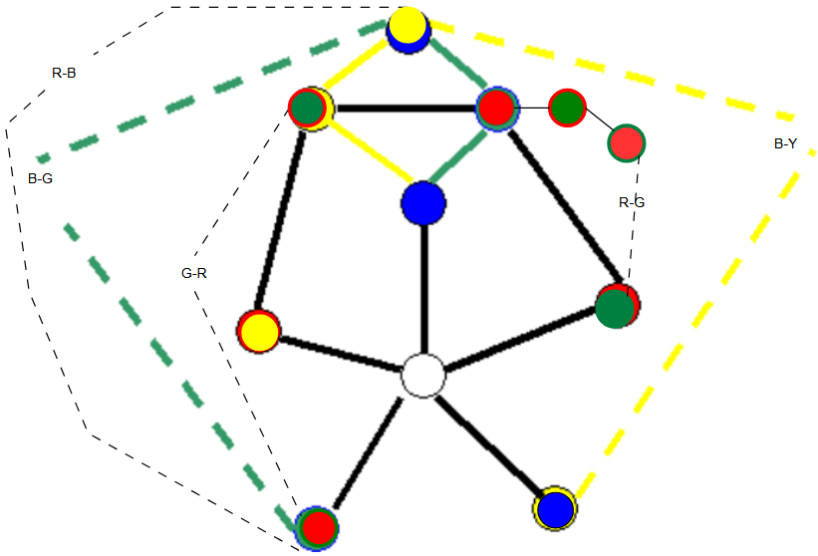
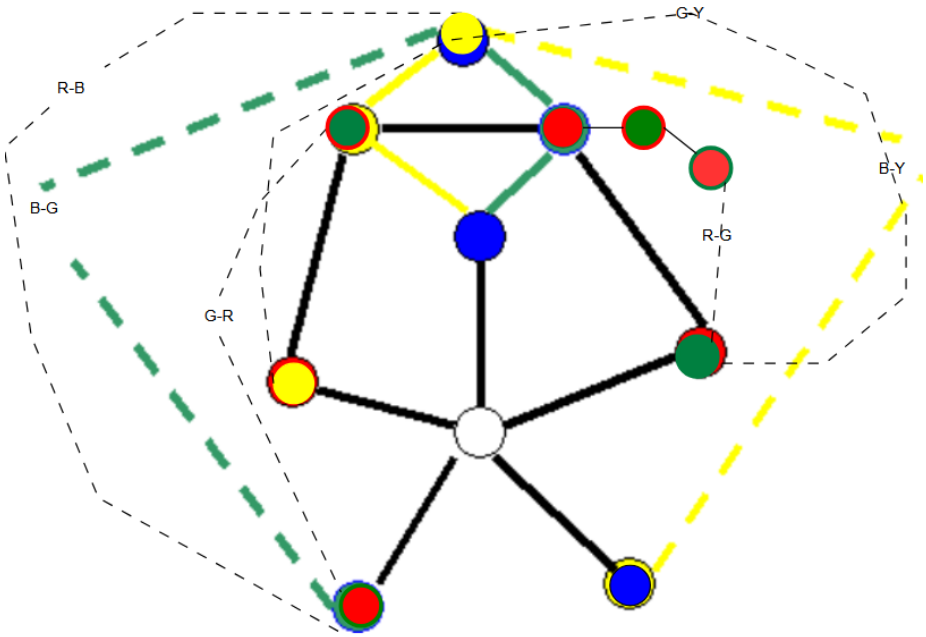
2) The second attempt is to change the double colouring, i.e. two yellows.

<u>The bottom right yellow can be toggled to blue.</u> There is no B-Y Kempe chain because the G-R Kempe chain on the left blocks off any possible B-Y Kempe chain. A B-Y Kempe chain can enter until blocked off by the G-R Kempe chain.



<u>The left yellow **can not** be toggled to green.</u> There can be a G-Y Kempe chain (added) as shown:



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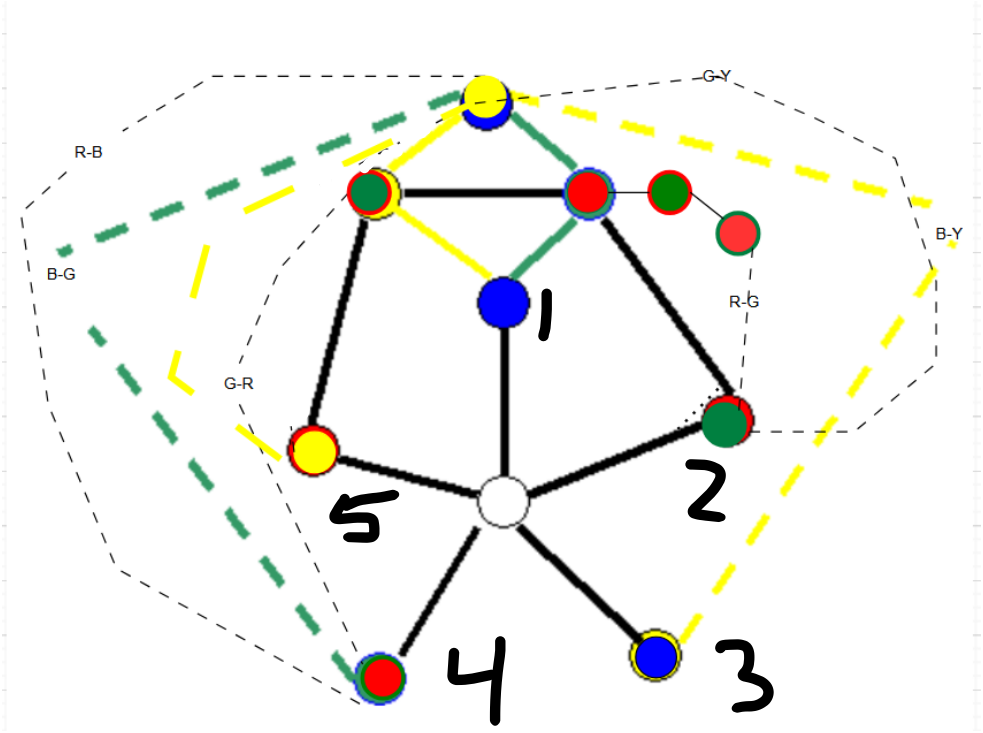
The G-Y Kempe chain is made clearer below. See the added yellow dotted line on the left. It goes through the Yellow-coloured node on top and through the (Y-)**G**-R Kempe chain on the left.



With this configuration (two blues, previously two yellows), there are:

1. G-Y Kempe chain (2-5)
2. R-G Kempe chain (2-4)
3. B-Y Kempe chain (3-5)

There is no B-R Kempe chain (1-4), due to the Y-node on top, but since there are 3 Kempe chains, any recolouring, will still result in 3 Kempe chains. Any recolouring will generate another double colouring with the same 3 Kempe chains that don't allow Kempe's method to work.

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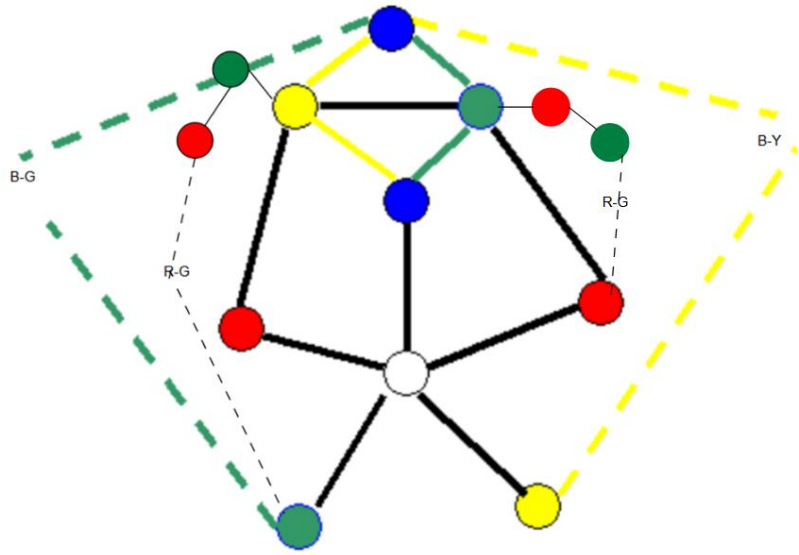
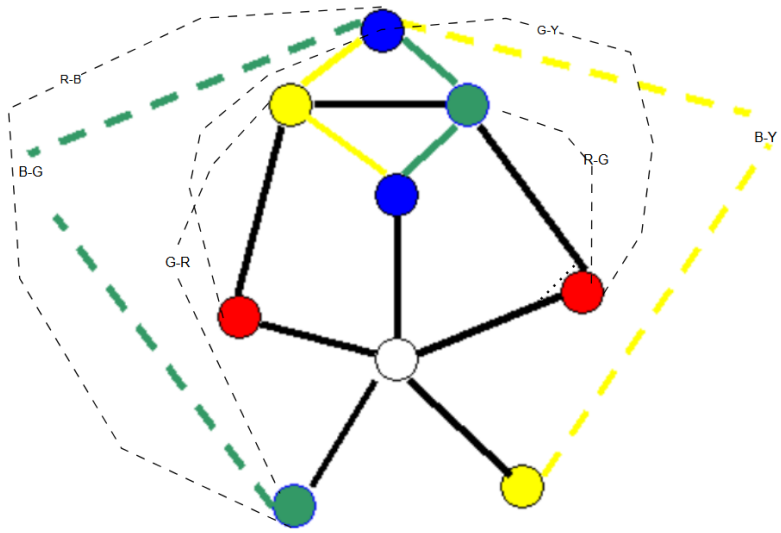
So the counterexample originates from a graph like this:



to form a graph like this (unofficial):